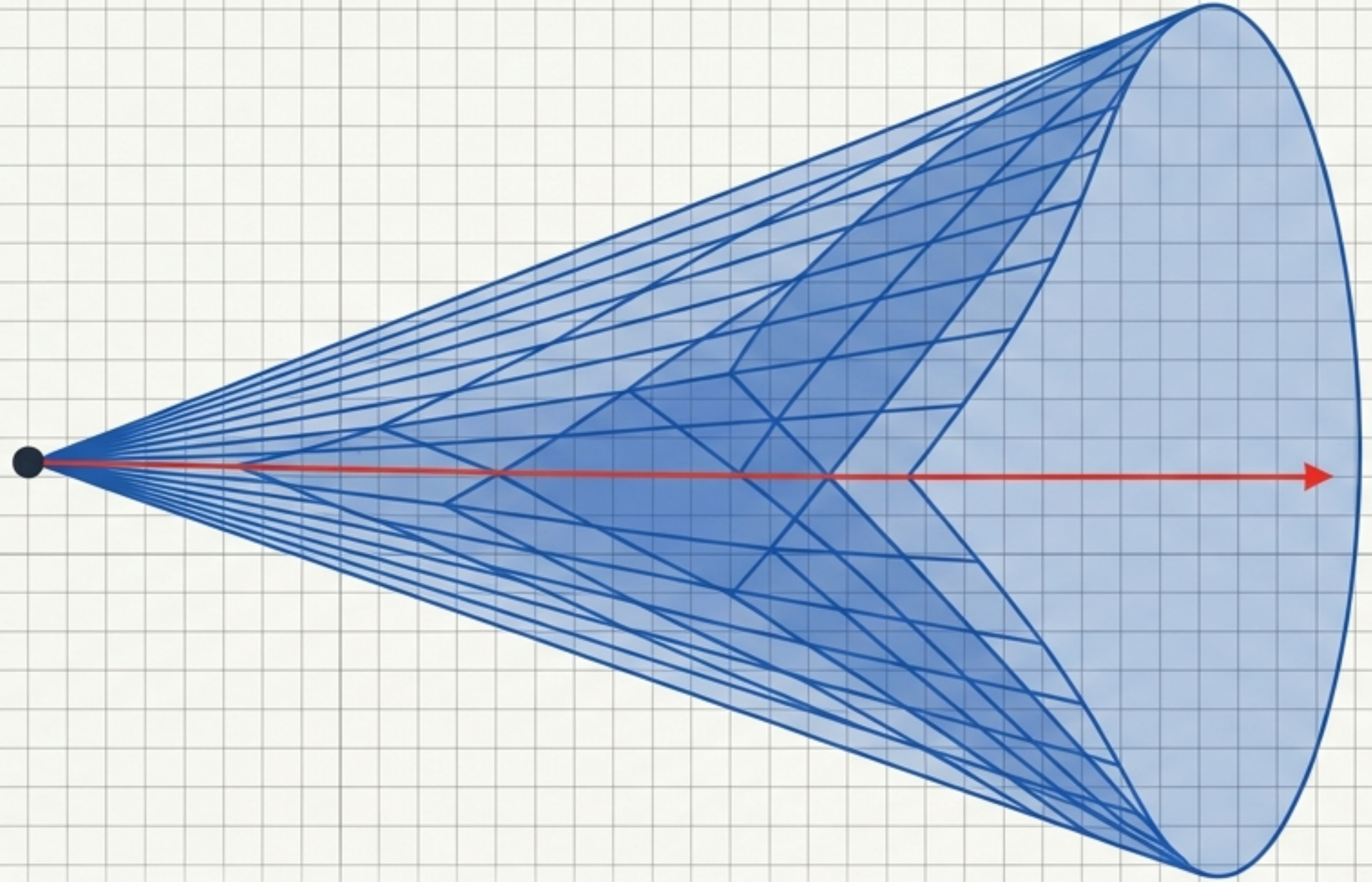




Continuation Geometry

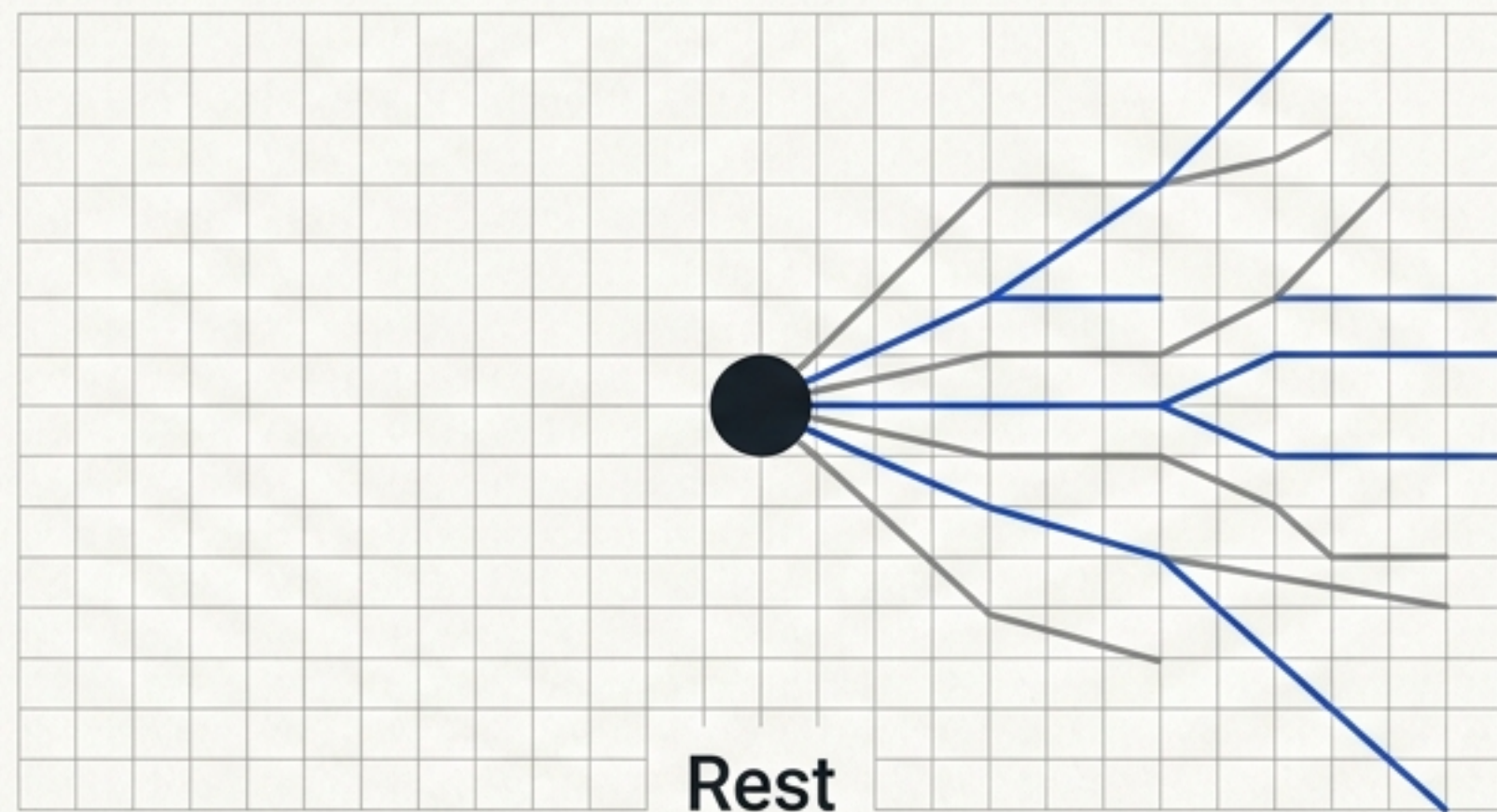
A system is defined not by its present state,
but by the width of what it can still become.

The Flaw of the Instantaneous Snapshot



Instantaneous State

Describes what the system is doing right now. Fails to distinguish between a room quiet because nothing needs to happen, and a room quiet because nothing can happen.



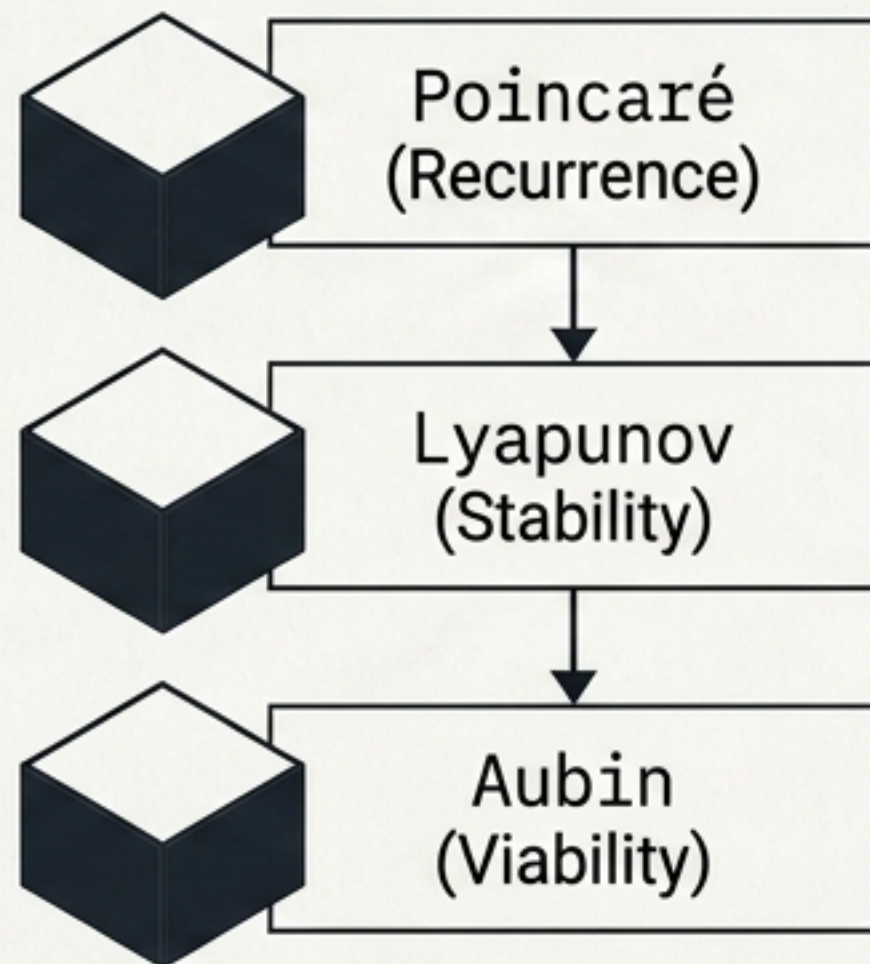
Continuation Space $\Gamma_K(x)$

Describes the whole reachable fan of futures. Captures the standing possibility of many outcomes, regardless of current velocity.

The Dual Ancestry of Continuation

The Persistence Lineage

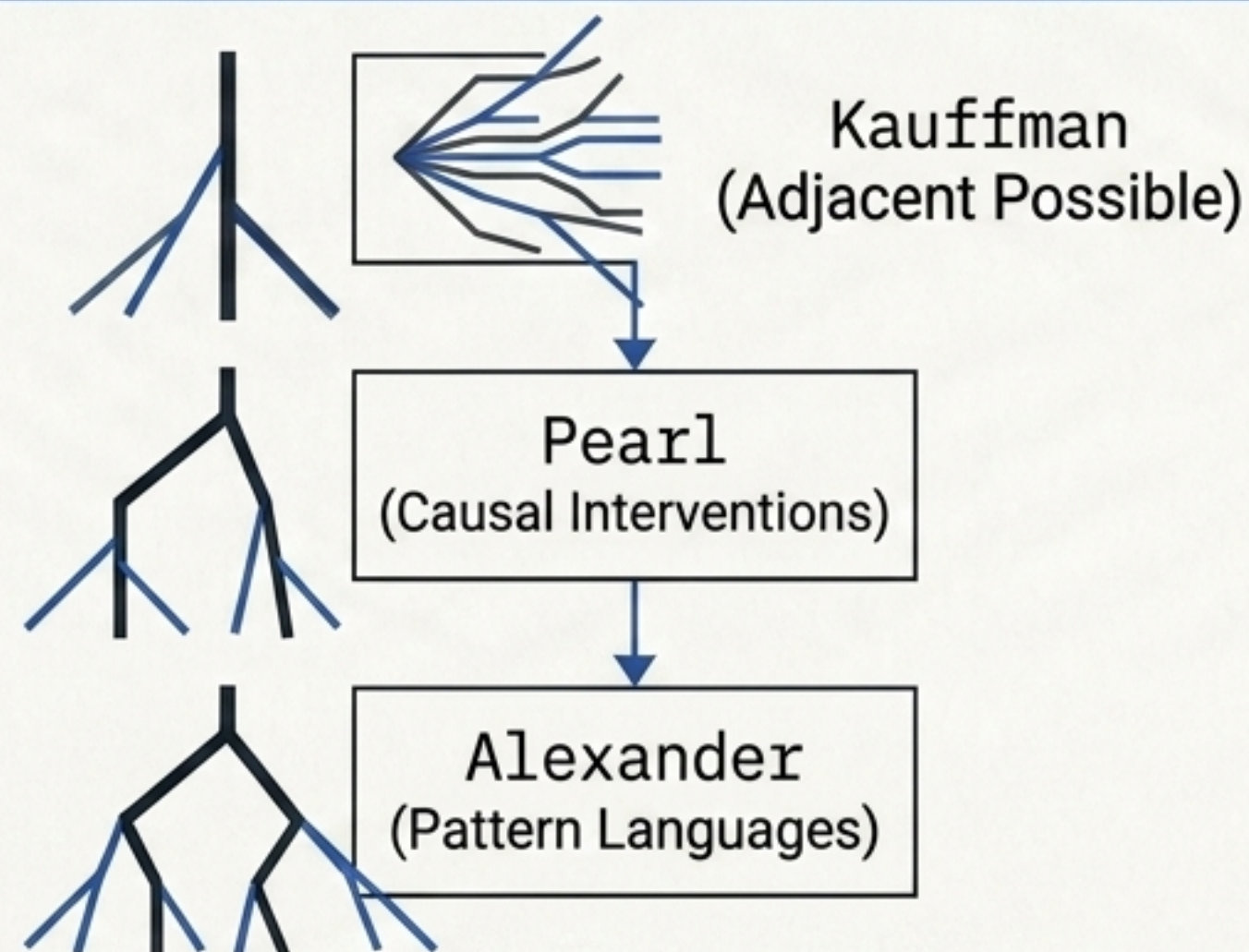
Core Question: How does a system keep going without breaking its own rules?



Focus: Existence of a future.

The Possibility-Space Lineage

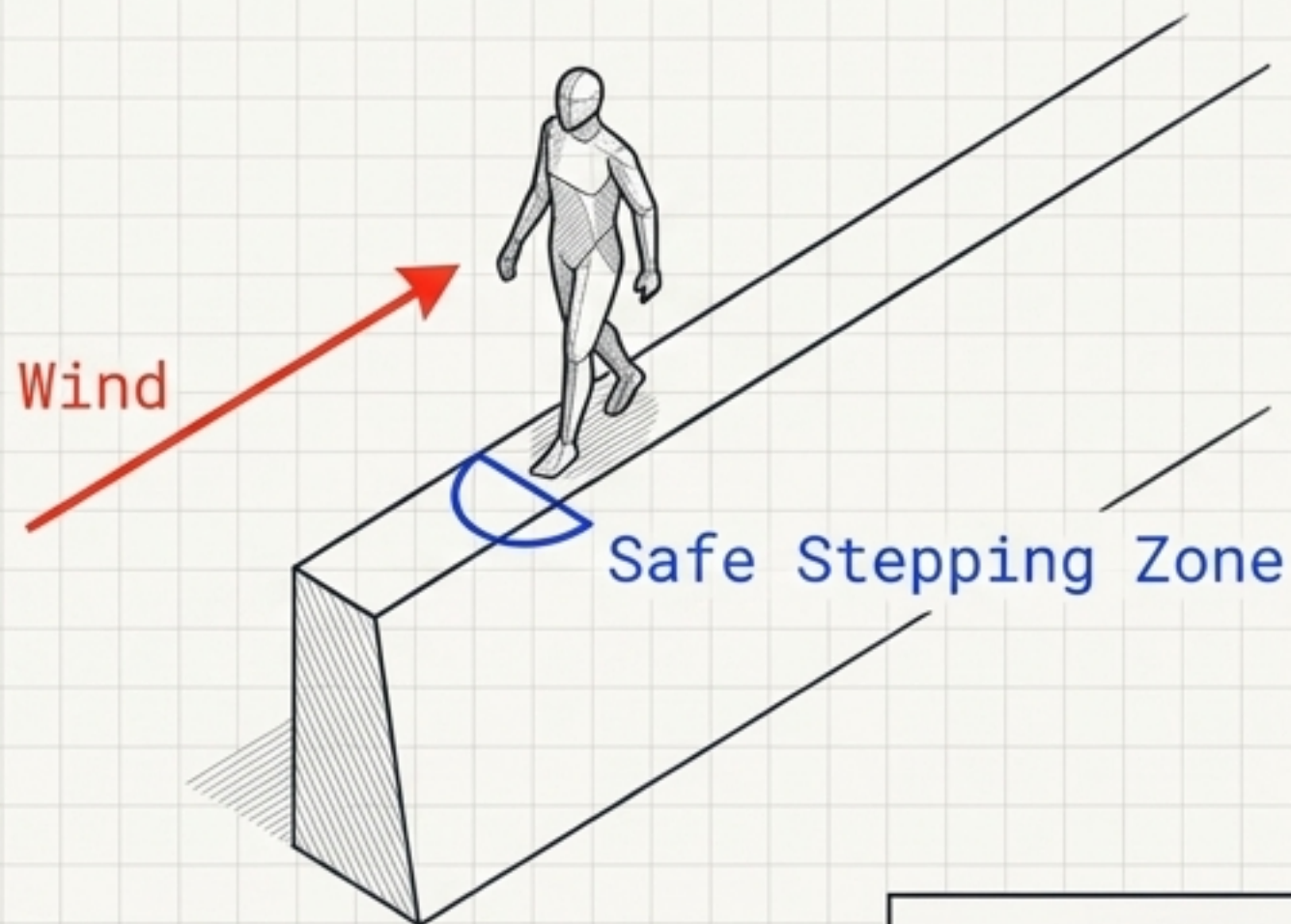
Core Question: What shape does the space of available futures take?



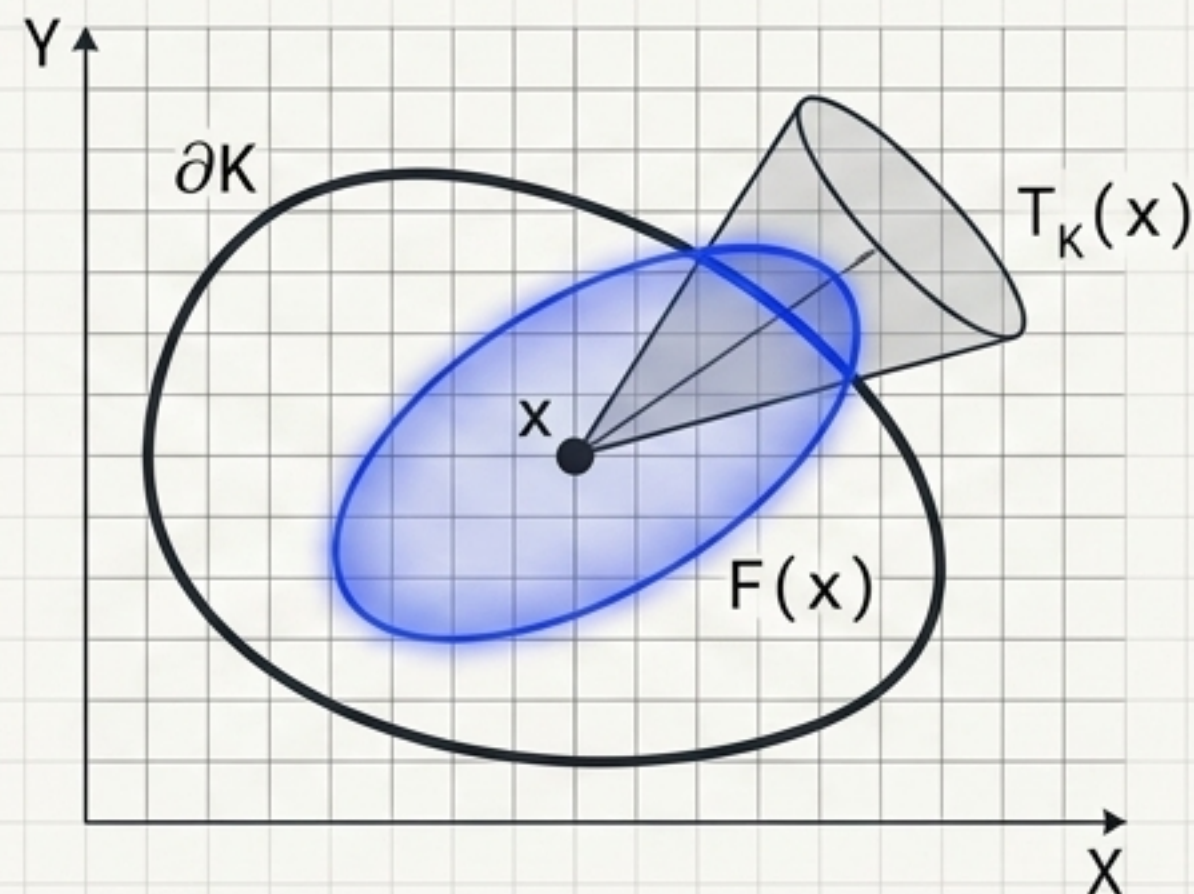
Focus: Options, growth, and structural layout.

The Local Test for Infinite Futures

The Metaphor



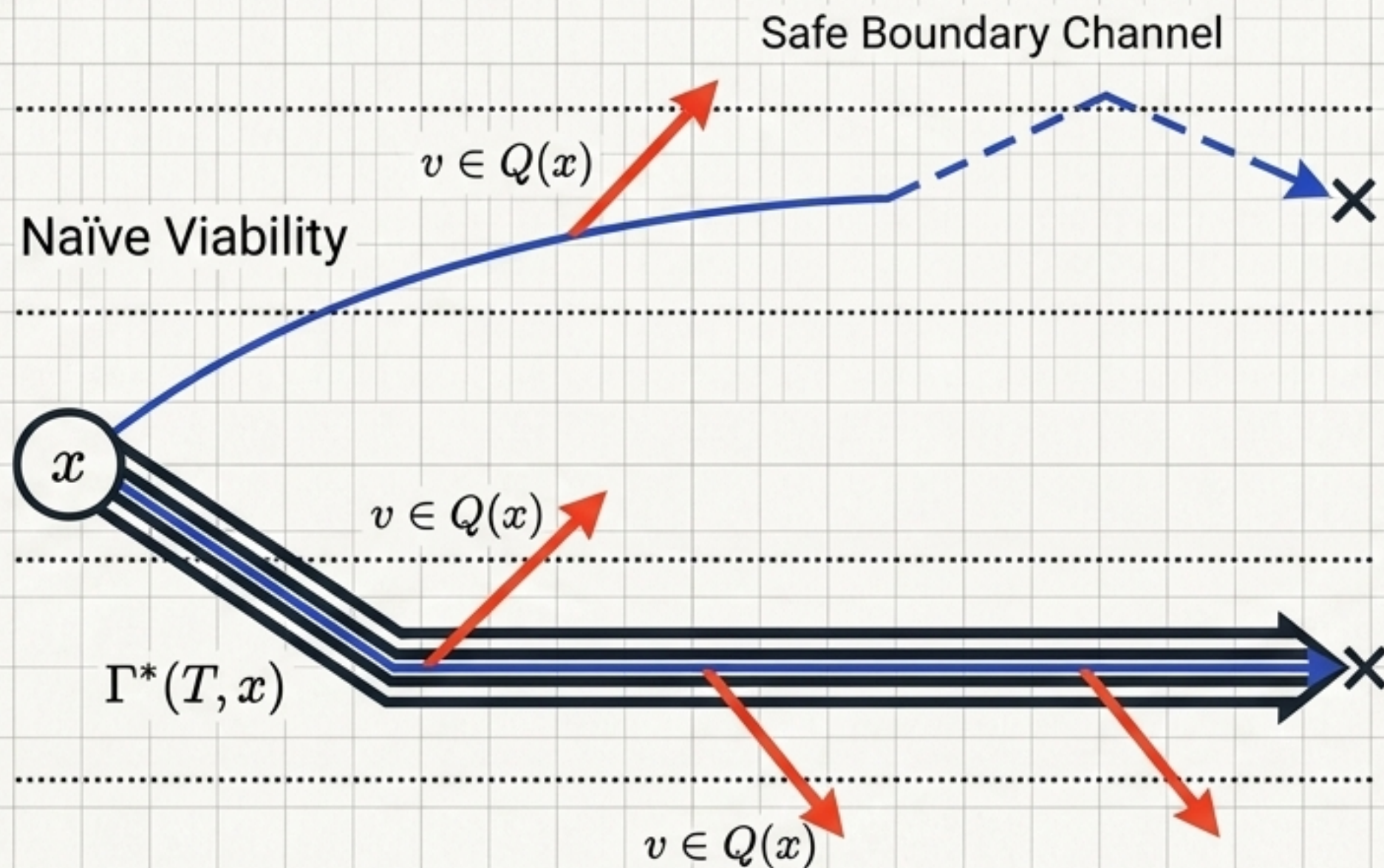
The Math



$$F(x) \cap T_K(x) \neq \emptyset$$

Takeaway: You don't need to simulate the whole future. You only need to know if the physics handed to you ($F(x)$) overlap with the directions that keep you inside the boundary ($T_K(x)$).

Surviving the Adversary



The Menu Γ

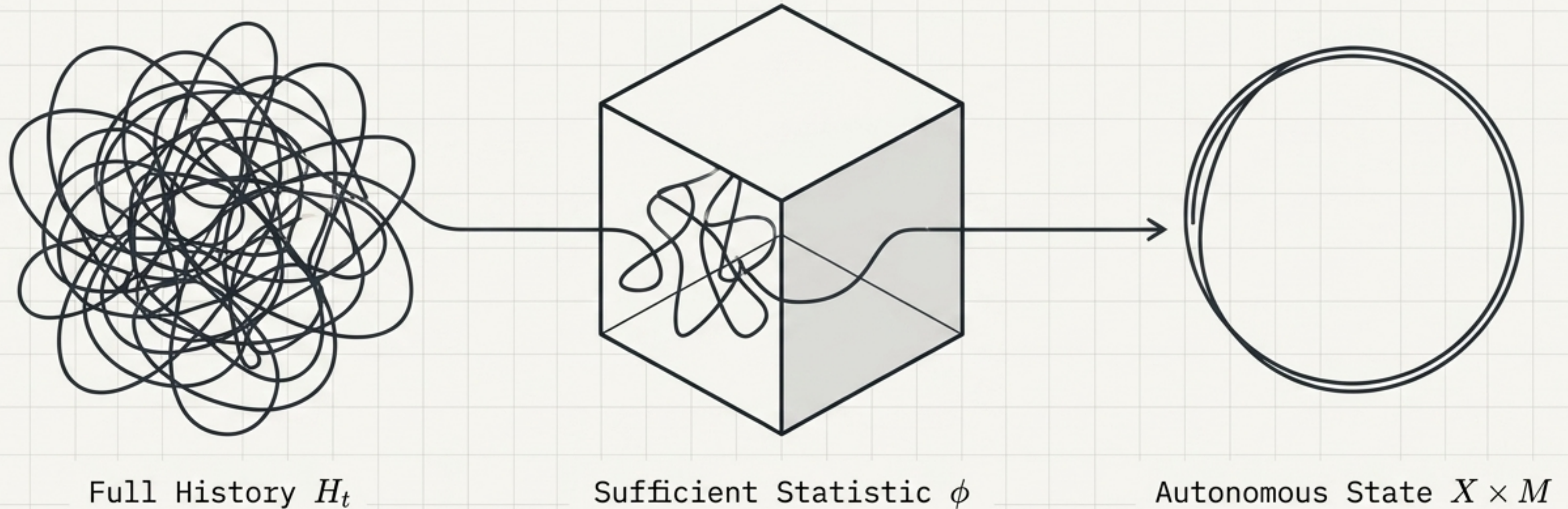
The full set of available controls $P(x)$ that keep the system viable in a vacuum.

The Selection Γ^*

The strictly smaller subset of controls that guarantee a defensible outcome even when the environment acts optimally against you.

Key Insight: Existence asks "Is there a way?"
Selection asks "Which way survives the worst plausible weather?"

The Impossibility of Perfect Memory



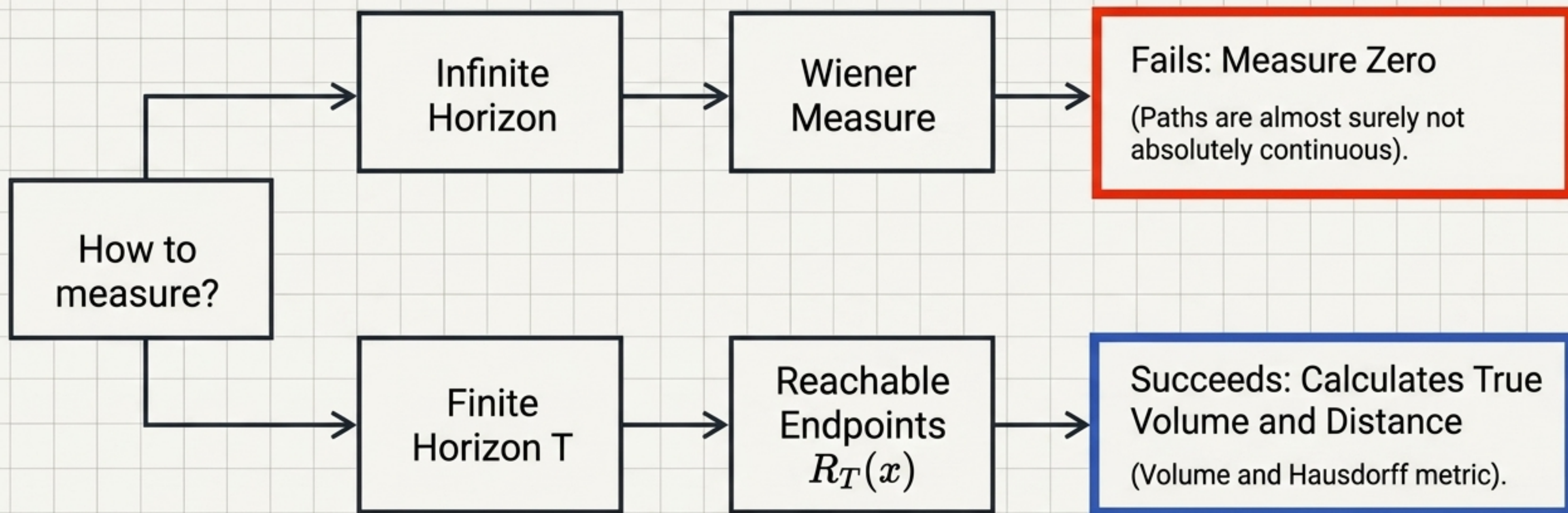
The Trap

A system can occupy identical states but face different futures based on invisible history (e.g., a biological scar, hidden technical debt).

The Limit Theorem

It is a mathematical impossibility to guarantee a finite-range sufficient statistic for all history-dependent systems. Reductions must be discovered case-by-case, not assumed.

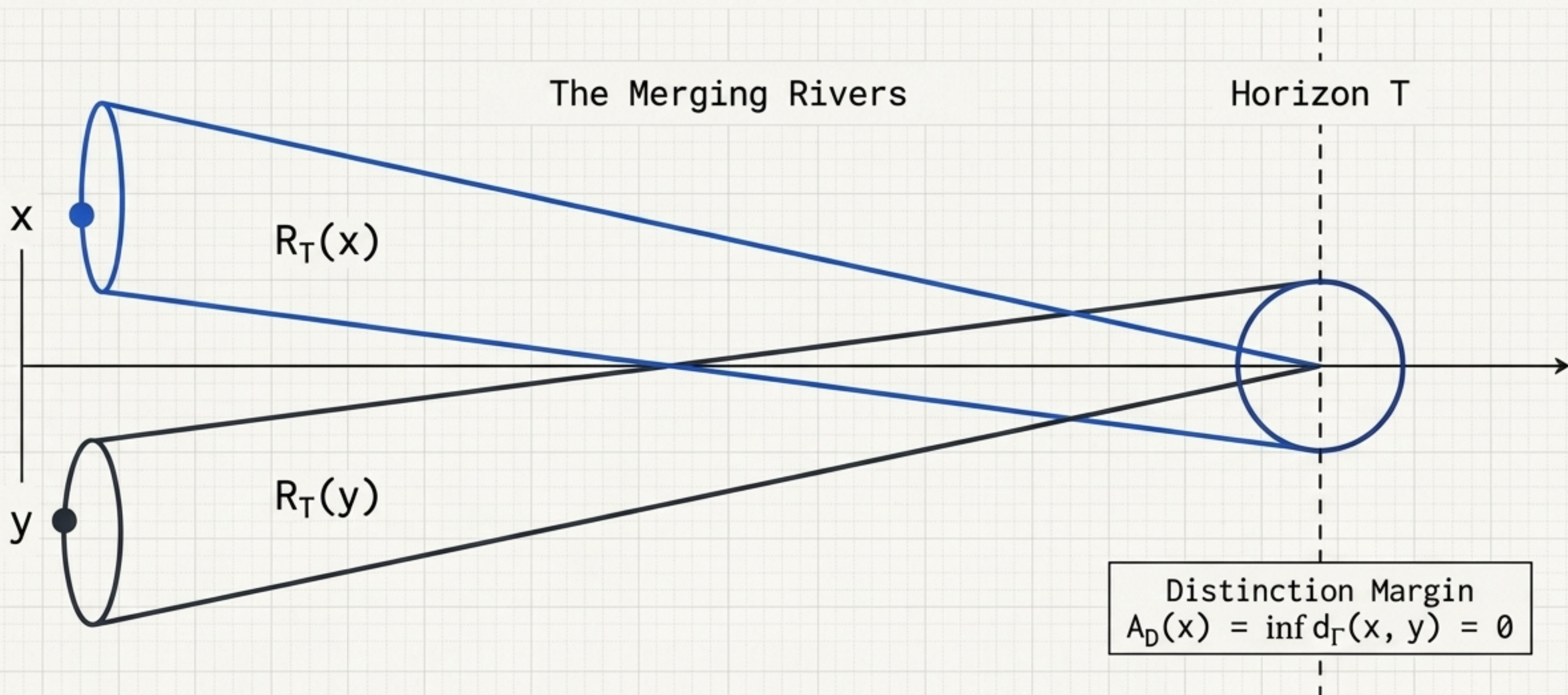
Measuring the Space of the Future



The Folding Warning

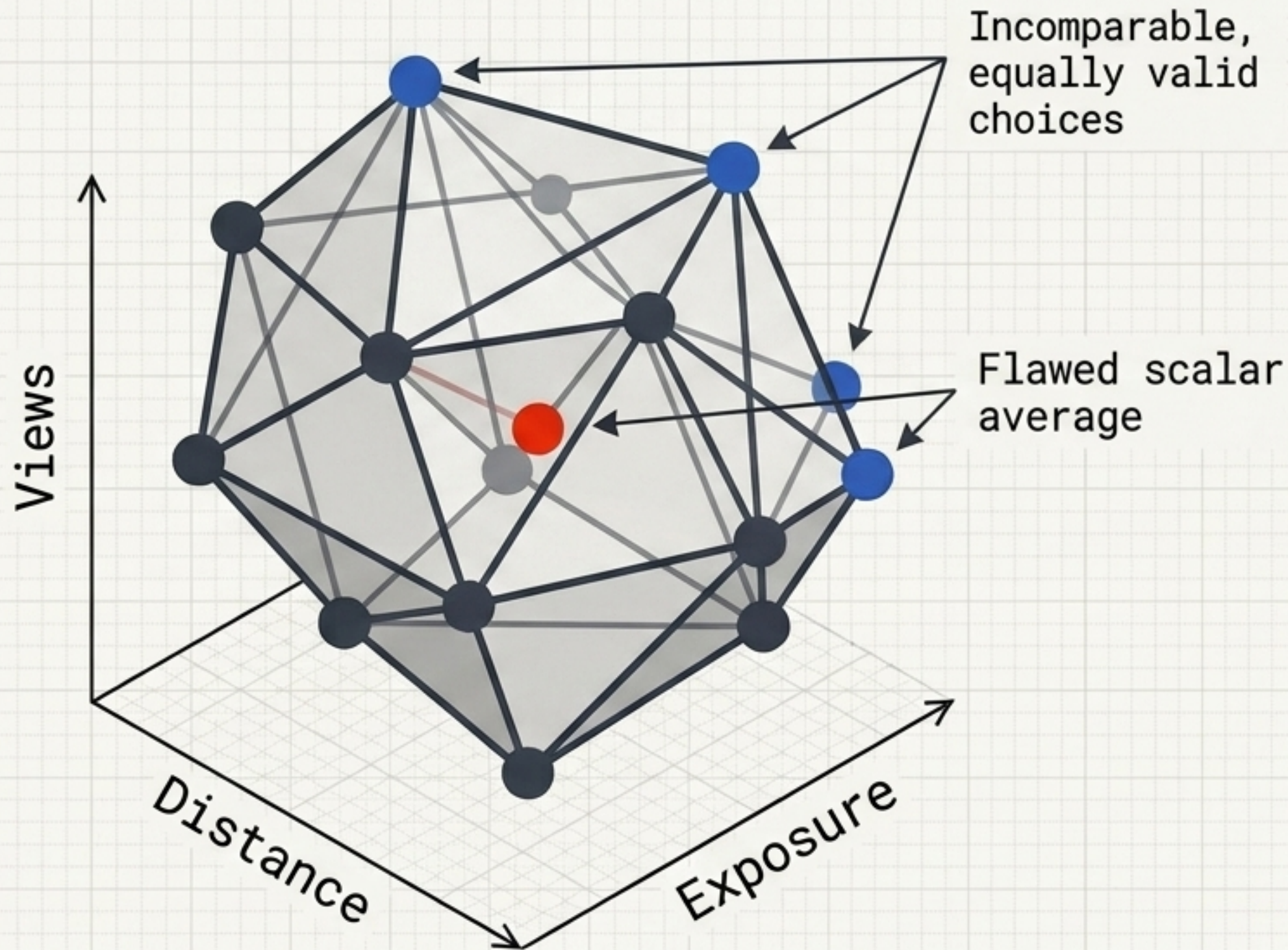
Reachable volume $\text{Vol}_T(x)$ is not strictly monotonic in time. Reachable sets can fold back on themselves, shrinking in volume even as time advances.

The Anatomy of a Distinction Collapse



Explanation: True repair restores the standing possibility of outcomes. Distinction collapse occurs when $A_D(x)$ hits zero. At this exact moment, the system “forgets” the difference between x and y . An observer looking only at the futures cannot tell the histories apart.

Resisting the Single Score



1. Dominance Relations

Weakest structure. Comparing criteria directly without numbers.

2. Pareto Structures

A continuation must be equal or better on every dimension to dominate. Leaves incomparable options intact.

3. The Last Resort

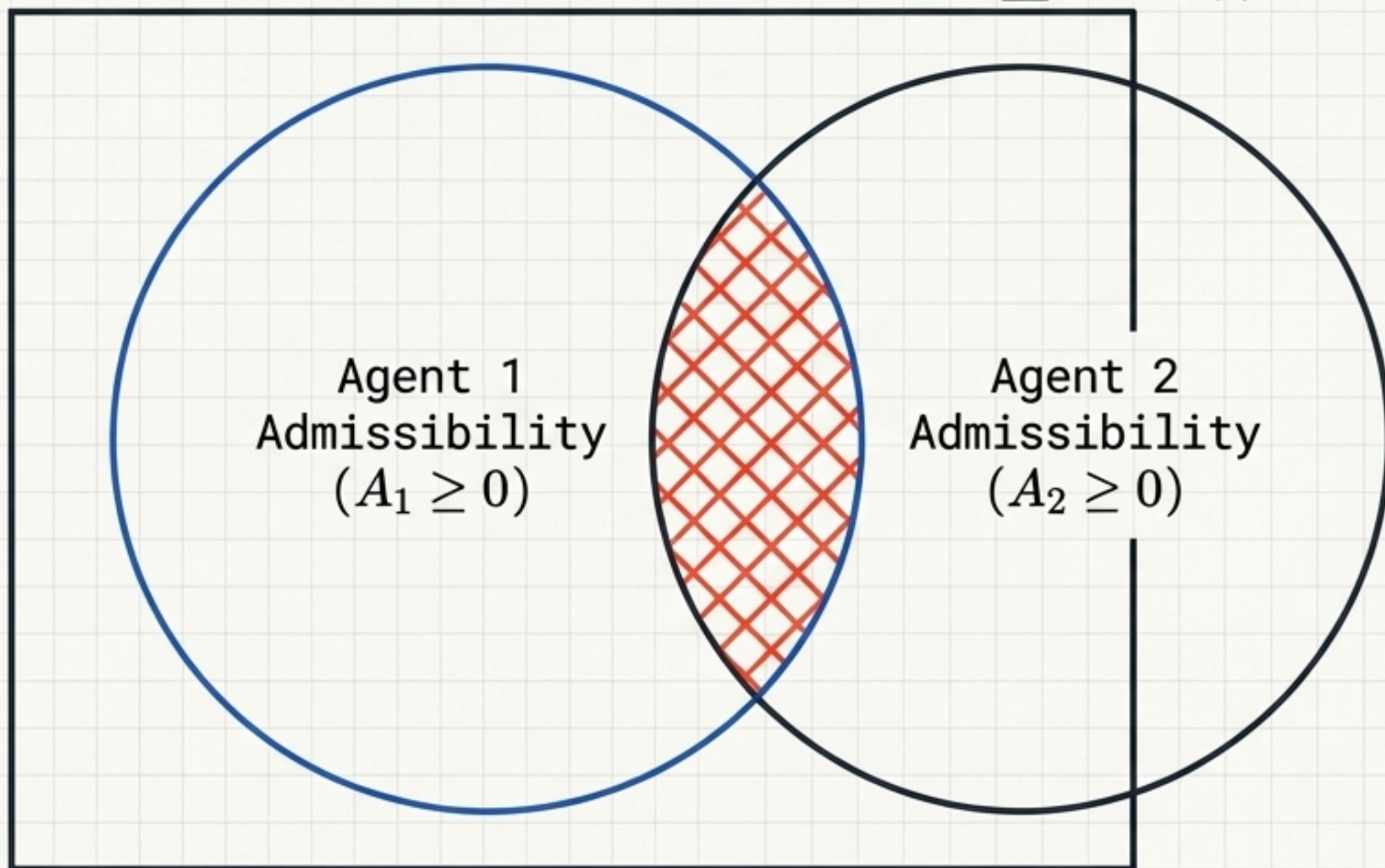
Collapsing criteria into a single scalar functional V .

Takeaway:

Averaging destroys critical constraint data. The scalar score should be the final, reluctant step, not the default.

When Shared Futures Collide

Grand Coalition Resource Limit: $\sum p_i \leq p(t)$



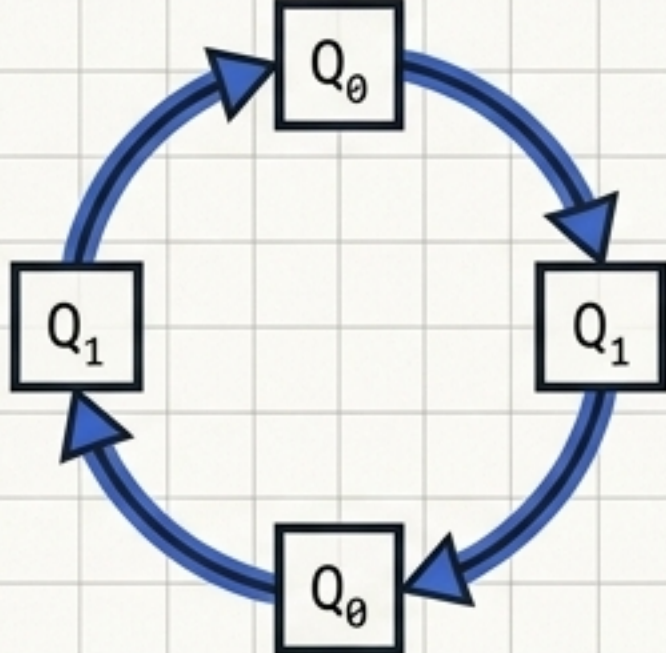
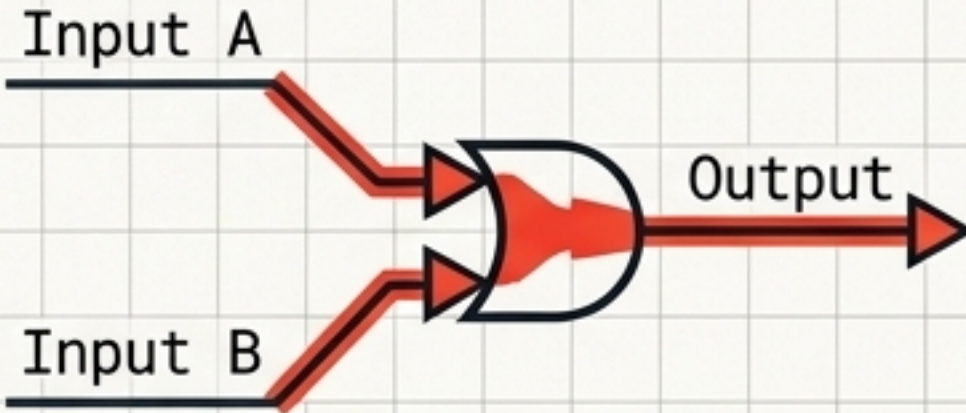
The Theorem

Individual rationality does not equal joint feasibility.

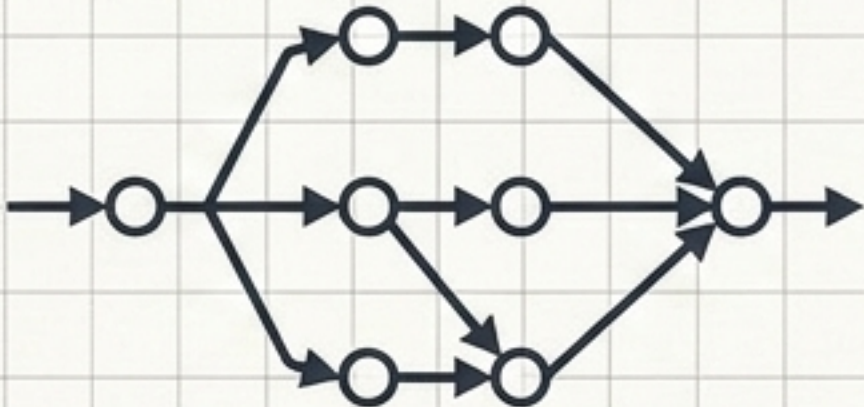
The Metaphor

Two ships in a channel. The state dynamics are shared, shared, but individual payoffs clash at the physical limit of the environment. The resource limit forces coordination failures even when the space itself is vast.

Testing the Theory on Small Machines

Combinational Logic (Half Adder)	Sequential Logic (2-Bit Counter)	8-Bit Byte (Traffic Light)
		
Volume = 1. A problem with no history and no continuation choice. The degenerate case.	Volume = 4. Saturating volume. Continuation with memory. The current count is the sufficient statistic.	The manifestation of Distinction Collapse . When a status bit is cleared to zero, distinct futures permanently merge. Hardware bugs are just forgotten futures.

Real-World Projections

Biology Target Morphologies	Software Technical Debt	Institutions Institutional Capture
 A regenerating organism's viable configurations (K) include every anatomy compatible with continued function. Bioelectric memory acts as the sufficient statistic.	 Debt is a shrinking continuation volume—fewer safe edits remain. A codebase regression is a literal distinction collapse where two states permanently merge.	 Reform is a capture basin problem. Coordination fails because shared resources (budget, public attention) cannot fulfill individual admissibility demands simultaneously.

The Limits of the Continuation Space

What $\Gamma(x)$ Organized Alone	Where $\Gamma(x)$ Required Outside Help
 Determining Viability and Existence.	 Defining the exact Resource Constraint.
 Calculating finite-horizon Volume and Distance.	 Identifying the Sufficient Statistic for history.
 Defining exact parameters for Distinction Collapse.	 Choosing the preferred Admissibility criteria.

Concluding Takeaway: A theory that maps exactly where it needs outside help is vastly **sturdier** than one that pretends it doesn't.